

YASHITA MOULI

Full-Stack Developer | Builder of Production-Grade AI Systems
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PROFILE

Detail-oriented B.Tech Electronics & Telecommunication student at NMIMS, Mumbai with hands-on experience in software development, data analytics, SQL database design, and AI-driven applications. Built and deployed multiple real-world projects ranging from scalable biometric intelligence systems and ML-based environmental platforms to productivity tools and dynamic AI applications. Strong foundation in Python, SQL, DBMS, and problem-solving, with a focus on building clean, scalable, and user-centric technical solutions. Seeking opportunities to contribute to high-impact software or data-driven teams while continuously learning and growing as an engineer.

FLAGSHIP PROJECT

NeuroSync - AI Behavioural Intelligence and Interview Analysis System | Multimodal AI 2026, Ongoing

Sole Developer | github.com/yashitamouli01-star/neurosync

- Architecting a real-time multimodal AI platform that fuses 3 ML engines: Vision Transformer (ViT) for facial micro-expression analysis via MediaPipe landmarks, Audio Spectrogram Transformer (AST) for vocal stress and pitch detection, and DeBERTa for semantic quality, filler-word analysis, and contradiction detection.
- Core research innovation: custom Cross-Attention Transformer fusion engine that combines face embeddings, voice embeddings, and text embeddings into unified behavioural intelligence, learning cross-modal relationships (does voice stress match facial emotion? does spoken confidence align with body language?).
- Optimised for CPU deployment via ONNX Runtime with INT8 quantisation; real-time pipeline streams webcam and microphone over WebSockets, Whisper transcribes speech live, dashboard updates with confidence graphs, hesitation zones, and eye contact metrics in under 100ms.

Stack: PyTorch · ViT · AST · DeBERTa · Whisper · MediaPipe · OpenCV · FastAPI · WebSockets · ONNX Runtime · INT8 Quantisation · Next.js · Supabase

FREELANCE PROJECTS

NewMarket.co.in - Hyperlocal Community Platform | Full-Stack SaaS

2026, Ongoing

Client Project | Personal Earnings: Rs. 10,000

- Built end-to-end bilingual (Hindi/English) platform integrating news, jobs, listings, and commerce; deployed on Vercel with 99.2% uptime over 3 months, 100K+ listings, 50K+ users.
- Designed normalised PostgreSQL schema with indexing and pagination achieving sub-200ms query response; JWT-based RBAC with 4 user tiers, admin dashboard, and automated flagging heuristics cutting manual moderation time by ~60%.
- Shipped nested comment threads, anonymous submissions, emoji reactions, i18n, and full SEO optimisation (open graph, sitemaps).

Stack: Next.js · TypeScript · Supabase (PostgreSQL) · Tailwind CSS · Vercel

The Business of Professionals - Peer Reputation Platform | Full-Stack MVP

2026, Upcoming

Client Project | Project Value: Rs. 50,000 | Personal Earnings: Rs. 22,000

- Scoped and architected full MVP for a professional credibility platform covering peer ratings, recruiter discovery, and AI-assisted integrity validation; moving beyond resume-based evaluation.
- Architecture: professional profiles, peer rating system, recruiter access dashboard, AI integrity layer, admin moderation, JWT auth; two-phase delivery: website MVP then mobile and enterprise scaling.

Stack: Next.js · Supabase · Render · Cloudflare · Razorpay · Vercel

PROJECT EXPERIENCE

ToxiScanAI - Drug Toxicity Prediction Platform | ML · Hackathon (CODECURE AI)

2024-2025

- End-to-end ML system predicting toxicity across 12 endpoints on ~12,000 compounds; 2,128-dim feature vectors, 36 models (RF, XGBoost, MLP), mean ROC-AUC 0.85; SMOTE for class imbalance, SHAP beeswarm plots for interpretability; Streamlit UI with SMILES input, Lipinski Ro5 validation, batch CSV inference.

Stack: Python · Scikit-learn · XGBoost · RDKit · SHAP · Streamlit · Plotly

BRI - Biometric Risk Intelligence System | UIDAI-Scale Data Engineering

2024-2025

- Engineered SQL database handling 1M+ biometric records; cut lookup query latency 35x (3.5s to 180ms) via targeted indexing; anomaly detection pipelines (Pandas) flagged duplicates across 5+ identity attributes; 3NF-normalised schemas with zero breaking schema changes.

Stack: Python · Pandas · MySQL · SQL · DBMS

- Silent Drift - Coastal Groundwater Intrusion Platform** | Hackathon · DU Hacks 5.0 2024
- AI decision-support platform predicting seawater intrusion affecting 10M+ people; 3D risk maps, ML forecasts (2027-2030), policy-ready reports, and Intrusion Risk Index (IRI) with RF regression and mitigation ROI calculator; built within a 26-hour sprint.
- Stack:** Python · Pandas · Scikit-learn · Plotly · PyDeck · ReportLab · Streamlit | *Demo:* youtu.be/TRTY2OYgHoM
- Focus Sense - Ethical Productivity Desktop App** | Personal Project 2024
- Desktop productivity tool with transparent focus-scoring, reflective analytics dashboards for distraction pattern identification; fully explainable UX, no gamification mechanics.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, C++

AI and ML: PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, ViT, AST, DeBERTa, Whisper, SHAP, ONNX Runtime, INT8 Quantisation, MediaPipe, OpenCV, RDKit

Backend and Data: FastAPI, WebSockets, REST APIs, PostgreSQL (Supabase), MySQL, JWT Auth, Pandas, NumPy, SQLAlchemy

Frontend: Next.js, React, TypeScript, Tailwind CSS, Framer Motion

DevOps and Tools: Git, Vercel, Render, GitHub Actions, Streamlit, Plotly, PyDeck

CS Fundamentals: Data Structures and Algorithms, OOP, DBMS, Normalisation, Query Optimisation

HACKATHONS

- DU Hacks 5.0 - Silent Drift (Coastal Groundwater Intrusion, AI/Sustainability)
- CODECURE AI - ToxiScanAI (Drug Toxicity Prediction, ML/Medical AI Track)
- ETHDenver 2026 - Blockchain/Web3 Development (Top 500 global participants)
- HyperBloom V2 | Google Build with AI | HACKANOVA 5.0 | Innov8 4.0 | Autho for AI Agents

LEADERSHIP AND EXPERIENCE

- Technical Member - Darwin Robotics Committee** | IGVC Team 2025 - Present
- Robotic system development for the Intelligent Ground Vehicle Competition (IGVC); motor driver integration, bot control, hardware testing, and cross-team troubleshooting with electronics and software teams.
- Executive - Student Member Development Committee** | IET MPSTME Aug 2024 - Jun 2025
- Coordinated technical events and workshops for a 100+ member student chapter over 11 months; managed scheduling, logistics, and member communications through structured planning templates.
- Junior Manager - Business Development** | AIESEC Aug 2024 - Feb 2025
- Drove outreach and partnership development for a global youth organisation; cross-functional campaign execution across Mumbai's student ecosystem.
- Volunteer - Drishti Foundation** | Social Work 2025
- Cleanliness drives, school and anganwadi teaching visits, plantation drives, and community awareness campaigns.

EDUCATION

- B.Tech - Electronics and Telecommunication Engineering** | NMIMS (MPSTME) Expected 2028
- Relevant Coursework: Data Structures, DBMS, Signals and Systems, OOP, Digital Electronics
- Class XII - Science (PCM)** | Narayana Institute 2022-2024

CERTIFICATIONS

- MySQL - Learning Education Hub (Dec 2025)
- Python with Data Science - Learning Education Hub (Feb 2025)
- Introduction to Programming Using Python - Udemy (Dec 2025)
- Python (Basic) - HackerRank (Apr 2025)
- Deloitte Australia Technology Job Simulation - Forage (Dec 2025)
- Electric Vehicle Powertrains Masterclass - MPSTME Industry Session